

Module 1: Fundamentals of Deep Learning and Medical Images

- Lecture 1: Research objects (tasks, cell detection, lesion detection), Intro to medical images: **X-ray, CT, MRI, and digital pathology**. Computer vision before deep learning: feature engineering. Why do we need deep learning(1 min)?
- Different tasks in medical image analysis, training task-specific models
- Lecture 2: Intro to neural networks.(15min)
- Lecture 3: and CNN (15min)

Module 2: Tasks in Medical Image Analysis

- Lecture : How to train a CNN? (code)
 - Lecture 1: Classification, RESNET
 - Lecture 2: Segmentation, UNET
 - Lecture 3: Synthesis, VAE, GAN

Module 3: Beyond task-specific model

- Self-supervised learning
 - Transformer
 - Diffusion
- Foundation model for analysis
 - Biomedpharse
 - sam2

Module 4: Clinical Integration, Challenges

- Challenges
 - Evaluation Challenges
 - Data privacy, federated learning
 - Skew-Distributed Data
 - Domain Adaptation
- Explanation, uncertainty

Lecture 4... (

<https://www.nature.com/articles/s41746-022-00592-y.pdf>

20 20 20 20... gpu heavy MSE certificate... lightweight...
Google colab gpu use..... Google colab pro

Recording...

Lecture 1.... CNN

Lecture 2...

Lightboard studio... (record it..... Lily Astone and Jake simpson)

Jake simpson video...

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Annotating ...

Look at the camera.....

No fine pattern ...

Solid color shirt....

Black shirt...belt and lapel....